In [5]:
```
!pip install -U transformers accelerate datasets soundfile librosa
faster-whisper jiwer
```

```
Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (5.0.0)
Collecting transformers
  Downloading transformers-5.4.0-py3-none-any.whl.metadata (32 kB)
Requirement already satisfied: accelerate in
/usr/local/lib/python3.12/dist-packages (1.13.0)
Requirement already satisfied: datasets in
/usr/local/lib/python3.12/dist-packages (4.0.0)
Collecting datasets
  Downloading datasets-4.8.4-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: soundfile in
/usr/local/lib/python3.12/dist-packages (0.13.1)
Requirement already satisfied: librosa in
/usr/local/lib/python3.12/dist-packages (0.11.0)
Collecting faster-whisper
  Downloading faster_whisper-1.2.1-py3-none-any.whl.metadata (16
kB)
Collecting jiwer
  Downloading jiwer-4.0.0-py3-none-any.whl.metadata (3.3 kB)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(1.7.1)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2.0.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(6.0.3)
Requirement already satisfied: regex>=2025.10.22 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(0.22.2)
Requirement already satisfied: typer in
/usr/local/lib/python3.12/dist-packages (from transformers)
(0.24.1)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(0.7.0)
Requirement already satisfied: tqdm>=4.27 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(4.67.3)
Requirement already satisfied: psutil in
/usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in
/usr/local/lib/python3.12/dist-packages (from accelerate)
(2.10.0+cu128)
```

```
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from datasets) (3.25.2)
Collecting pyarrow>=21.0.0 (from datasets)
  Downloading pyarrow-23.0.1-cp312-cp312-
manylinux_2_28_x86_64.whl.metadata (3.1 kB)
Requirement already satisfied: dill<0.4.2,>=0.3.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: httpx<1.0.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (0.28.1)
Requirement already satisfied: xxhash in
/usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.20 in
/usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2026.2.0,>=2023.1.0 in
/usr/local/lib/python3.12/dist-packages (from fsspec[http]
<=2026.2.0,>=2023.1.0->datasets) (2025.3.0)
Requirement already satisfied: cffi>=1.0 in
/usr/local/lib/python3.12/dist-packages (from soundfile) (2.0.0)
Requirement already satisfied: audioread>=2.1.9 in
/usr/local/lib/python3.12/dist-packages (from librosa) (3.1.0)
Requirement already satisfied: numba>=0.51.0 in
/usr/local/lib/python3.12/dist-packages (from librosa) (0.60.0)
Requirement already satisfied: scipy>=1.6.0 in
/usr/local/lib/python3.12/dist-packages (from librosa) (1.16.3)
Requirement already satisfied: scikit-learn>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from librosa) (1.6.1)
Requirement already satisfied: joblib>=1.0 in
/usr/local/lib/python3.12/dist-packages (from librosa) (1.5.3)
Requirement already satisfied: decorator>=4.3.0 in
/usr/local/lib/python3.12/dist-packages (from librosa) (4.4.2)
Requirement already satisfied: pooch>=1.1 in
/usr/local/lib/python3.12/dist-packages (from librosa) (1.9.0)
Requirement already satisfied: soxr>=0.3.2 in
/usr/local/lib/python3.12/dist-packages (from librosa) (1.0.0)
Requirement already satisfied: typing_extensions>=4.1.1 in
/usr/local/lib/python3.12/dist-packages (from librosa) (4.15.0)
Requirement already satisfied: lazy_loader>=0.1 in
/usr/local/lib/python3.12/dist-packages (from librosa) (0.5)
Requirement already satisfied: msgpack>=1.0 in
/usr/local/lib/python3.12/dist-packages (from librosa) (1.1.2)
Collecting ctranslate2<5,>=4.0 (from faster-whisper)
  Downloading ctranslate2-4.7.1-cp312-cp312-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (10 kB)
Collecting onnxruntime<2,>=1.14 (from faster-whisper)
  Downloading onnxruntime-1.24.4-cp312-cp312-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.2 kB)
Collecting av>=11 (from faster-whisper)
  Downloading av-17.0.0-cp311-abi3-
manylinux_2_28_x86_64.whl.metadata (4.6 kB)
Requirement already satisfied: click>=8.1.8 in
/usr/local/lib/python3.12/dist-packages (from jiwer) (8.3.1)
Collecting rapidfuzz>=3.9.7 (from jiwer)
  Downloading rapidfuzz-3.14.3-cp312-cp312-
```

```
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (12 kB)
Requirement already satisfied: pycparser in
/usr/local/lib/python3.12/dist-packages (from cffi>=1.0-
>soundfile) (3.0)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from ctranslate2<5,>=4.0-
>faster-whisper) (75.2.0)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in
/usr/local/lib/python3.12/dist-packages (from fsspec[http]
<=2026.2.0,>=2023.1.0->datasets) (3.13.3)
Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx<1.0.0-
>datasets) (4.12.1)
Requirement already satisfied: certifi in
/usr/local/lib/python3.12/dist-packages (from httpx<1.0.0-
>datasets) (2026.2.25)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1.0.0-
>datasets) (1.0.9)
Requirement already satisfied: idna in
/usr/local/lib/python3.12/dist-packages (from httpx<1.0.0-
>datasets) (3.11)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1.0.0->datasets) (0.16.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.5.0->transformers) (1.4.2)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from numba>=0.51.0-
>librosa) (0.43.0)
Requirement already satisfied: flatbuffers in
/usr/local/lib/python3.12/dist-packages (from
onnxruntime<2,>=1.14->faster-whisper) (25.12.19)
Requirement already satisfied: protobuf in
/usr/local/lib/python3.12/dist-packages (from
onnxruntime<2,>=1.14->faster-whisper) (5.29.6)
Requirement already satisfied: sympy in
/usr/local/lib/python3.12/dist-packages (from
onnxruntime<2,>=1.14->faster-whisper) (1.14.0)
Requirement already satisfied: platformdirs>=2.5.0 in
/usr/local/lib/python3.12/dist-packages (from pooch>=1.1->librosa)
(4.9.4)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (3.4.6)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (2.5.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.1.0-
>librosa) (3.6.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
```

```
>accelerate) (3.1.6)
Requirement already satisfied: cuda-bindings==12.9.4 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.93)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90
in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.90)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.90)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (11.3.3.83)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.5.8.93)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (12.8.93)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (1.13.1.3)
Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.12/dist-packages (from torch>=2.0.0-
>accelerate) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-
bindings==12.9.4->torch>=2.0.0->accelerate) (1.4.3)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
```

```
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
(2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
(2025.3)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->transformers)
(1.5.4)
Requirement already satisfied: rich>=12.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->transformers)
(13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer->transformers)
(0.0.4)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0-
>datasets) (2.6.1)
Requirement already satisfied: aiosignal>=1.4.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0-
>datasets) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0-
>datasets) (25.4.0)
Requirement already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0-
>datasets) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0-
>datasets) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0-
>datasets) (0.4.1)
Requirement already satisfied: yarl<2.0,>=1.17.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0-
>datasets) (1.23.0)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from python-
dateutil>=2.8.2->pandas->datasets) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
>transformers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
>transformers) (2.19.2)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy-
>onnxruntime<2,>=1.14->faster-whisper) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
```

```
/usr/local/lib/python3.12/dist-packages (from jinja2-
>torch>=2.0.0->accelerate) (3.0.3)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-
py>=2.2.0->rich>=12.3.0->typer->transformers) (0.1.2)
Downloading transformers-5.4.0-py3-none-any.whl (10.1 MB)
[2K   [90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m
[32m10.1/10.1 MB[0m [31m91.8 MB/s[0m eta [36m0:00:00[0m
[?25hDownloading datasets-4.8.4-py3-none-any.whl (526 kB)
[2K   [90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m
[32m527.0/527.0 kB[0m [31m49.2 MB/s[0m eta [36m0:00:00[0m
[?25hDownloading faster_whisper-1.2.1-py3-none-any.whl (1.1 MB)
[2K   [90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m
[32m1.1/1.1 MB[0m [31m77.6 MB/s[0m eta [36m0:00:00[0m
[?25hDownloading jiwer-4.0.0-py3-none-any.whl (23 kB)
Downloading av-17.0.0-cp311-abi3-manylinux_2_28_x86_64.whl (36.4
MB)
[2K   [90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m
[32m36.4/36.4 MB[0m [31m29.7 MB/s[0m eta [36m0:00:00[0m
[?25hDownloading ctranslate2-4.7.1-cp312-cp312-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (39.0 MB)
[2K   [90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m
[32m39.0/39.0 MB[0m [31m23.6 MB/s[0m eta [36m0:00:00[0m
[?25hDownloading onnxruntime-1.24.4-cp312-cp312-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (17.2 MB)
[2K   [90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m
[32m17.2/17.2 MB[0m [31m105.7 MB/s[0m eta [36m0:00:00[0m
[?25hDownloading pyarrow-23.0.1-cp312-cp312-
manylinux_2_28_x86_64.whl (47.6 MB)
[2K   [90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m
[32m47.6/47.6 MB[0m [31m14.7 MB/s[0m eta [36m0:00:00[0m
[?25hDownloading rapidfuzz-3.14.3-cp312-cp312-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (3.2 MB)
[2K   [90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m
[32m3.2/3.2 MB[0m [31m120.4 MB/s[0m eta [36m0:00:00[0m
[?25hInstalling collected packages: rapidfuzz, pyarrow,
ctranslate2, av, onnxruntime, jiwer, datasets, transformers,
faster-whisper
  Attempting uninstall: pyarrow
    Found existing installation: pyarrow 18.1.0
    Uninstalling pyarrow-18.1.0:
      Successfully uninstalled pyarrow-18.1.0
  Attempting uninstall: datasets
    Found existing installation: datasets 4.0.0
    Uninstalling datasets-4.0.0:
      Successfully uninstalled datasets-4.0.0
  Attempting uninstall: transformers
    Found existing installation: transformers 5.0.0
    Uninstalling transformers-5.0.0:
      Successfully uninstalled transformers-5.0.0
Successfully installed av-17.0.0 ctranslate2-4.7.1 datasets-4.8.4
faster-whisper-1.2.1 jiwer-4.0.0 onnxruntime-1.24.4 pyarrow-23.0.1
rapidfuzz-3.14.3 transformers-5.4.0
```

In [6]:
```
import pandas as pd
from jiwer import wer, cer
import time
import re

audios = [
    {
        "file": "test_en_1.wav",
        "text": "okay let's go let's start our testing",
        "lang": "en"
    },
    {
        "file": "test_en_2.wav",
        "text": "now im going to the university so i can work on my
project",
        "lang": "en"
    },
    {
        "file": "test_en_3.wav",
        "text": "the weather today is sunny and very hot to be honest",
        "lang": "en"
    },
    {
        "file": "test_en_4.wav",
        "text": "im really tired i just wanna go home and sleep",
        "lang": "en"
    },
    {
        "file": "test_en_5.wav",
        "text": "hello what do you think who is the best singer alive",
        "lang": "en"
    },
    {
        "file": "test_en_6.wav",
        "text": "hey siri how to find the purpose of life",
        "lang": "en"
    },
    # ---------- ru
    {
        "file": "test_ru_1.wav",
        "text": "всем привет сегодня у нас погода отличная",
        "lang": "ru"
    },
    {
        "file": "test_ru_2.wav",
        "text": "сегодня у нас наурыз и народа тут очень много",
        "lang": "ru"
    },
    {
        "file": "test_ru_3.wav",
        "text": "когда мы покупали впервые машину там такой приятный
запах",
        "lang": "ru"
    },
    {
        "file": "test_ru_4.wav",
        "text": "я люблю гулять ночью потому что ночью обычно тихо на
```

```
улице",
        "lang": "ru"
    },
    {
        "file": "test_ru_5.wav",
        "text": "привет алиса поставь пожалуйста будильник на семь
утра",
        "lang": "ru"
    },
    {
        "file": "test_ru_6.wav",
        "text": "привет а какова вероятность что лифт упадет",
        "lang": "ru"
    },
    # ---------- mix
    {
        "file": "test_mix_1.wav",
        "text": "вчера у нас была презентация i guess it was cringe",
        "lang": "mix"
    },
    {
        "file": "test_mix_2.wav",
        "text": "we planned to go to swimming но там было занято",
        "lang": "mix"
    },
    {
        "file": "test_mix_3.wav",
        "text": "я смотрел the first spider man и мне это очень сильно
понравилось",
        "lang": "mix"
    },
    {
        "file": "test_mix_4.wav",
        "text": "hello hello what will be the погода завтра",
        "lang": "mix"
    },
    {
        "file": "test_mix_5.wav",
        "text": "это мне кажется или one plus one is really three",
        "lang": "mix"
    },
    {
        "file": "test_mix_6.wav",
        "text": "с днём рождения тебя i wish you all the best",
        "lang": "mix"
    },
]

results = []

def basic_clean(text):
    text = text.lower()
    text = re.sub(r'[^\w\s]', '', text)
    return " ".join(text.split())

def record_result(model_name, audio, pred, duration):
    audio["text"] = basic_clean(audio["text"])
```

```
    prediction = basic_clean(pred)

    results.append({
        "Model": model_name,
        "File": audio["file"],
        "Language": audio["lang"],
        "Real": audio["text"],
        "Predicted": prediction.strip(),
        "WER": round(wer(audio["text"], prediction), 4),
        "CER": round(cer(audio["text"], prediction), 4),
        "Time_Sec": round(duration, 2)
    })
```

In [7]:
```
import torch
from transformers import pipeline

model1 = pipeline(
    "automatic-speech-recognition",
    model="openai/whisper-large-v3",
    torch_dtype=torch.float16,
    device="cuda"
)

for audio in audios:
    start = time.time()
    output = model1(audio["file"])
    record_result("Whisper-v3", audio, output["text"], time.time() -
start)

del model1
torch.cuda.empty_cache()
```

```
/usr/local/lib/python3.12/dist-
packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in
your settings tab (https://huggingface.co/settings/tokens), set
it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still
optional to access public models or datasets.
  warnings.warn(
```

```
config.json: 0.00B [00:00, ?B/s]
```

```
`torch_dtype` is deprecated! Use `dtype` instead!
Warning: You are sending unauthenticated requests to the HF
Hub. Please set a HF_TOKEN to enable higher rate limits and
faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN
to enable higher rate limits and faster downloads.
```

```
model.safetensors:   0%|          | 0.00/3.09G [00:00<?, ?B/s]
```

```
Loading weights:   0%|          | 0/1259 [00:00<?, ?it/s]
```

```
generation_config.json: 0.00B [00:00, ?B/s]
```

```
tokenizer_config.json: 0.00B [00:00, ?B/s]
```

```
vocab.json: 0.00B [00:00, ?B/s]
```

```
tokenizer.json: 0.00B [00:00, ?B/s]

merges.txt: 0.00B [00:00, ?B/s]

normalizer.json: 0.00B [00:00, ?B/s]

added_tokens.json: 0.00B [00:00, ?B/s]

special_tokens_map.json: 0.00B [00:00, ?B/s]

preprocessor_config.json:   0%|          | 0.00/340 [00:00<?, ?
B/s]
```

```
Using custom `forced_decoder_ids` from the (generation) config.
This is deprecated in favor of the `task` and `language`
flags/config options.
Transcription using a multilingual Whisper will default to
language detection followed by transcription instead of
translation to English. This might be a breaking change for
your use case. If you want to instead always translate your
audio to English, make sure to pass `language='en'`. See
https://github.com/huggingface/transformers/pull/28687 for more
details.
A custom logits processor of type <class
'transformers.generation.logits_process.SuppressTokensLogitsPro
cessor'> has been passed to `.generate()`, but it was also
created in `.generate()`, given its parameterization. The
custom <class
'transformers.generation.logits_process.SuppressTokensLogitsPro
cessor'> will take precedence. Please check the docstring of
<class
'transformers.generation.logits_process.SuppressTokensLogitsPro
cessor'> to see related `.generate()` flags.
A custom logits processor of type <class
'transformers.generation.logits_process.SuppressTokensAtBeginLo
gitsProcessor'> has been passed to `.generate()`, but it was
also created in `.generate()`, given its parameterization. The
custom <class
'transformers.generation.logits_process.SuppressTokensAtBeginLo
gitsProcessor'> will take precedence. Please check the
docstring of <class
'transformers.generation.logits_process.SuppressTokensAtBeginLo
gitsProcessor'> to see related `.generate()` flags.
You seem to be using the pipelines sequentially on GPU. In
order to maximize efficiency please use a dataset
```

In [8]:
```
from faster_whisper import WhisperModel
import time

model2 = WhisperModel("large-v3", device="cuda", compute_type="float16")

for audio in audios:
    start = time.time()

    segments, info = model2.transcribe(audio["file"], beam_size=1)

    pred = "".join([segment.text for segment in segments])

    record_result("Faster-Whisper-V3", audio, pred, time.time() - start)

del model2
torch.cuda.empty_cache()
```

In [9]:
```
import torch
from transformers import pipeline

model3 = pipeline(
    "automatic-speech-recognition",
    model="openai/whisper-large-v3-turbo",
    torch_dtype=torch.float16,
    device="cuda"
)

for audio in audios:
    start = time.time()
    result = model3(audio["file"])
    record_result("Whisper-Turbo", audio, result["text"], time.time() -
start)

del model3
torch.cuda.empty_cache()
```

```
config.json: 0.00B [00:00, ?B/s]

model.safetensors:   0%|          | 0.00/1.62G [00:00<?, ?B/s]

Loading weights:   0%|          | 0/587 [00:00<?, ?it/s]

generation_config.json: 0.00B [00:00, ?B/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]

vocab.json: 0.00B [00:00, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

merges.txt: 0.00B [00:00, ?B/s]

normalizer.json: 0.00B [00:00, ?B/s]

added_tokens.json: 0.00B [00:00, ?B/s]

special_tokens_map.json: 0.00B [00:00, ?B/s]

preprocessor_config.json:   0%|          | 0.00/340 [00:00<?, ?
B/s]
```

dalwe rezy

In [10]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame(results)

plt.figure(figsize=(10, 6))
sns.set_style("whitegrid")
plot = sns.barplot(data=df, x="Model", y="WER", hue="Language",
palette="viridis")
plt.title("ASR Performance: Word Error Rate (Lower is Better)")
plt.ylabel("WER (0.0 - 1.0)")
plt.show()

df_sorted = df.sort_values(by=["WER", "CER", "Time_Sec"], ascending=
[True, True, True])

display(df_sorted[["Model", "Language", "WER", "CER", "Time_Sec",
"Predicted"]].reset_index(drop=True))
```

	Model	Language	WER	CER	Time_Sec	Predicted
0	Whisper-Turbo	ru	0.0000	0.0000	0.54	всем привет сегодня у нас погода отличная
1	Whisper-Turbo	en	0.0000	0.0000	0.59	now im going to the university so i can work o...
2	Whisper-Turbo	en	0.0000	0.0000	0.59	the weather today is sunny and very hot to be ...
3	Whisper-Turbo	ru	0.0000	0.0000	0.60	привет а какова

	Model	Language	WER	CER	Time_Sec	Predicted
						вероятность что лифт упадет
4	Whisper-Turbo	ru	0.0000	0.0000	0.63	я люблю гулять ночью потому что ночью обычно т...
5	Whisper-Turbo	mix	0.0000	0.0000	0.64	вчера у нас была презентация i guess it was cr...
6	Whisper-v3	ru	0.0000	0.0000	1.08	всем привет сегодня у нас погода отличная
7	Whisper-v3	en	0.0000	0.0000	1.18	the weather today is sunny and very hot to be ...
8	Whisper-v3	mix	0.0000	0.0000	1.20	это мне кажется или one plus one is really three
9	Faster-Whisper-V3	ru	0.0000	0.0000	1.21	всем привет сегодня у нас погода отличная
10	Faster-Whisper-V3	en	0.0000	0.0000	1.24	the weather today is sunny and very hot to be ...
11	Faster-Whisper-V3	en	0.0000	0.0000	1.28	hello what do you think who is the best singer...
12	Faster-Whisper-V3	mix	0.0000	0.0000	1.29	я смотрел the first spider man и мне это очень...
13	Whisper-v3	en	0.0000	0.0000	1.35	now im going to the university so i can work o...

	Model	Language	WER	CER	Time_Sec	Predicted
14	Whisper-v3	ru	0.0000	0.0000	1.43	привет а какова вероятность что лифт упадет
15	Whisper-v3	en	0.0000	0.0000	1.53	hello what do you think who is the best singer...
16	Whisper-v3	ru	0.0000	0.0000	1.53	я люблю гулять ночью потому что ночью обычно т...
17	Faster-Whisper-V3	ru	0.0000	0.0000	1.60	я люблю гулять ночью потому что ночью обычно т...
18	Faster-Whisper-V3	ru	0.0000	0.0000	1.62	привет а какова вероятность что лифт упадет
19	Whisper-Turbo	en	0.0000	0.0000	1.74	okay lets go lets start our testing
20	Faster-Whisper-V3	en	0.0000	0.0000	3.17	okay lets go lets start our testing
21	Whisper-v3	en	0.0000	0.0000	4.11	okay lets go lets start our testing
22	Whisper-Turbo	mix	0.1000	0.0227	0.88	с днем рождения тебя i wish you all the best
23	Whisper-Turbo	mix	0.1000	0.0417	0.56	это мне кажется или one plus one is really free

	Model	Language	WER	CER	Time_Sec	Predicted
24	Whisper-Turbo	ru	0.1111	0.0444	0.58	сегодня у нас наврез и народа тут очень много
25	Faster-Whisper-V3	ru	0.1111	0.0444	1.21	сегодня у нас наврез и народа тут очень много
26	Whisper-v3	en	0.1111	0.0500	1.17	hi siri how to find the purpose of life
27	Faster-Whisper-V3	en	0.1111	0.0500	1.24	hi siri how to find the purpose of life
28	Whisper-Turbo	ru	0.1111	0.0702	0.62	когда мы покупали впервые машину там был такой...
29	Faster-Whisper-V3	ru	0.1111	0.0702	1.29	когда мы покупали впервые машину там был такой...
30	Whisper-v3	ru	0.1111	0.0702	1.48	когда мы покупали впервые машину там был такой...
31	Whisper-Turbo	ru	0.1250	0.0741	0.62	привет алиса поставь пожалуйста будильник на 7...
32	Whisper-v3	ru	0.1250	0.0741	1.53	привет алиса поставь пожалуйста будильник на 7...
33	Faster-Whisper-V3	ru	0.1250	0.0741	1.57	привет алиса поставь пожалуйста будильник на 7...

	Model	Language	WER	CER	Time_Sec	Predicted
34	Faster-Whisper-V3	en	0.1538	0.0345	1.38	now i am going to the university so i can work...
35	Whisper-Turbo	mix	0.1667	0.0154	0.61	я смотрел the first spiderman и мне это очень ...
36	Whisper-v3	mix	0.1667	0.0154	1.36	я смотрел the first spiderman и мне это очень ...
37	Whisper-Turbo	en	0.2000	0.0870	0.68	im really tired i just want to go home and sleep
38	Faster-Whisper-V3	en	0.2000	0.0870	1.22	im really tired i just want to go home and sleep
39	Whisper-v3	en	0.2000	0.0870	1.34	im really tired i just want to go home and sleep
40	Whisper-v3	ru	0.2222	0.0667	1.17	сегодня у нас наврез и народу тут очень много
41	Whisper-v3	mix	0.2500	0.3571	1.09	hello hello what will be the weather tomorrow
42	Faster-Whisper-V3	mix	0.2500	0.3571	1.27	hello hello what will be the weather tomorrow
43	Faster-Whisper-V3	mix	0.3000	0.1818	1.09	с днём рождения тебя а ещё и all the best
44	Whisper-Turbo	en	0.3333	0.1750	0.64	hi sri how to find the proposal of life

	Model	Language	WER	CER	Time_Sec	Predicted
45	Faster-Whisper-V3	mix	0.5000	0.3600	1.64	вчера у нас была презентация а я здесь вот хрень
46	Whisper-v3	mix	0.5000	0.3600	1.72	вчера у нас была презентация а я сидел скрынч
47	Whisper-v3	mix	0.6000	0.2727	1.43	с днём рождения тебя а я ищу altebest
48	Whisper-Turbo	mix	0.6000	0.5319	0.63	мы планируем идти на спирт но там было занято
49	Faster-Whisper-V3	mix	0.6000	0.5319	1.31	мы планируем пойти на воду но там было занято
50	Whisper-v3	mix	0.6000	0.5319	1.81	мы планируем пойти на воду но там было занято
51	Faster-Whisper-V3	mix	0.6000	0.5417	1.27	это мне кажется или 1 плюс 1 это действительно 3
52	Whisper-Turbo	mix	0.7500	0.6190	0.56	привет привет как будет погода завтра
53	Whisper-Turbo	en	1.0000	0.8846	0.76	привет что вы думаете кто лучший сцентр в жизни

In [11]:
```
import pandas as pd

model_summary = (
    df.groupby("Model")
      .agg({
          "WER": "mean",
          "CER": "mean",
          "Time_Sec": "mean"
      })
      .reset_index()
      .sort_values("WER")
)

print(model_summary)
```

```
                  Model       WER       CER  Time_Sec
2         Whisper-v3  0.160339  0.104728  1.528333
0  Faster-Whisper-V3  0.170117  0.129594  1.438889
1      Whisper-Turbo  0.199844  0.142556  0.692778
```

In [12]:
```
wer_lang = (
    df.groupby(["Model", "Language"])["WER"]
      .mean()
      .reset_index()
)

wer_pivot = wer_lang.pivot(index="Model", columns="Language",
values="WER")

print(wer_pivot)
```

```
Language                 en       mix        ru
Model
Faster-Whisper-V3  0.077483  0.375000  0.057867
Whisper-Turbo      0.255550  0.286117  0.057867
Whisper-v3         0.051850  0.352783  0.076383
```

In [13]:
```
latency = (
    df.groupby("Model")["Time_Sec"]
      .mean()
      .reset_index()
      .sort_values("Time_Sec")
)

print(latency)
```

```
                  Model  Time_Sec
1      Whisper-Turbo  0.692778
0  Faster-Whisper-V3  1.438889
2         Whisper-v3  1.528333
```

In [15]:
```
comparison = (
    df.groupby("Model")["WER"]
      .describe()
)

print(comparison)
```

```
                      count      mean       std  min  25%      50%
75%  max
Model
Faster-Whisper-V3   18.0  0.170117  0.205497  0.0  0.0  0.11110
0.237500  0.6
Whisper-Turbo       18.0  0.199844  0.291326  0.0  0.0  0.10555
0.191675  1.0
Whisper-v3          18.0  0.160339  0.206827  0.0  0.0  0.11110
0.216650  0.6
```

In [14]:
```
hardest = df.sort_values("WER", ascending=False).head(10)

print(hardest[["Model", "Language", "WER", "Predicted"]])
```

```
                Model Language     WER  \
40      Whisper-Turbo       en  1.0000
51      Whisper-Turbo      mix  0.7500
31  Faster-Whisper-V3      mix  0.6000
34  Faster-Whisper-V3      mix  0.6000
17         Whisper-v3      mix  0.6000
13         Whisper-v3      mix  0.6000
49      Whisper-Turbo      mix  0.6000
12         Whisper-v3      mix  0.5000
30  Faster-Whisper-V3      mix  0.5000
41      Whisper-Turbo       en  0.3333

                                          Predicted
40   привет что вы думаете кто лучший сцентр в жизни
51             привет привет как будет погода завтра
31     мы планируем пойти на воду но там было занято
34  это мне кажется или 1 плюс 1 это действительно 3
17             с днём рождения тебя а я ищу altebest
13     мы планируем пойти на воду но там было занято
49     мы планируем идти на спирт но там было занято
12     вчера у нас была презентация а я сидел скрынч
30  вчера у нас была презентация а я здесь вот хрень
41           hi sri how to find the proposal of life
```

trash dalwe

In [1]:
```
# !pip install jiwer
# from jiwer import wer, cer
```

```
Collecting jiwer
  Downloading jiwer-4.0.0-py3-none-any.whl.metadata (3.3 kB)
Requirement already satisfied: click>=8.1.8 in
/usr/local/lib/python3.12/dist-packages (from jiwer) (8.3.1)
Collecting rapidfuzz>=3.9.7 (from jiwer)
  Downloading rapidfuzz-3.14.3-cp312-cp312-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (12 kB)
Downloading jiwer-4.0.0-py3-none-any.whl (23 kB)
Downloading rapidfuzz-3.14.3-cp312-cp312-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (3.2 MB)
[2K   [90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m
[32m3.2/3.2 MB[0m [31m52.4 MB/s[0m eta [36m0:00:00[0m
[?25hInstalling collected packages: rapidfuzz, jiwer
Successfully installed jiwer-4.0.0 rapidfuzz-3.14.3
```

In [3]:
```
# org = "hello my name is baurzhan im from kazakhstan"
# pred = "hello my name is pao zhang im from kazakhstan"
# print(round(wer(org, pred), 4))
```

```
0.25
```

In [1]:
```
# from nemo.collections.asr.models import EncDecMultiTaskModel

# canary_model = EncDecMultiTaskModel.from_pretrained('nvidia/canary-
1b')
# canary_model.to("cuda")

# for audio in audios:
#     start_time = time.time()
#     res = canary_model.transcribe(
#         paths2audio_files=[audio["file"]],
#         task="asr",
#         source_lang=audio["lang"],
#         batch_size=1
#     )

#     prediction = res[0] if isinstance(res, list) else res

#     record_result("Canary-1B", audio, prediction, time.time() -
start_time)

# del canary_model
# torch.cuda.empty_cache()
```

In [ ]:

Exported with runcell — convert notebooks to HTML or PDF anytime at runcell.dev.